QR Code for
report verification

Case ID	:	**2600127567**	**Sample Type**	:	**FFPE TISSUE BLOCK**
Name	:	**MR. KADIRVEL (OQG2603250662)**	**Date & Time Collected**	:	**25-Mar-2026 12:40 PM**
Sex/Age	:	**Male/65 Years**	**Date & Time Received**	:	**06-Apr-2026 05:56 PM**
Bill. Loc.	:	**OncQuest Laboratories Ltd., New Delhi**	**Date & Time Reported**	:	**21-Apr-2026 06:02 PM**
Ref. By	:		**Report Version**	:	**1**
Indication	:				

Solidseq Comprehensive Panel
On Illumina Novaseq 6000 Platform

Clinical Indication:
Case of Carcinoma Esophagus
Block No.: 1627/25-C

Positive
(Clinically significant Variant and CNV detected)

Report Highlights:

Tier I	Tier II	Tier III	MSI	Therapies	Clinical Trials
0	2	0	Stable	0	20

Key Findings:

Tier	Gene & Transcript	Variant	Exon	Coverage/VAF
IIC	APC NM_000038.6	c.1312+3A>G (Splice Site Mutation)	10	1037x / 13.51%
IIC	TP53 NM_000546.6	c.524G>A p.(R175H)	5	1080x / 5.00%

CNV Type	Genes	Copy Number
Amplification	MYC	6

**Genetic test results are reported based on the recommendations of AMP-ASCO-CAP guidelines.*
***Public data sources included in relevant targeted therapies, guidelines: EMA, FDA, ESMO, NCCN.*

Multi-Gene Biomarkers Results:

Assay Biomarker	Result	Interpretation
Microsatellite Instability (MSI)	Stable	Stable Microsatellite detected

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 # Correlation of the genetic findings with the clinical condition of the patient is required to arrive at accurate diagnosis, prognosis or therapeutic decisions.

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Clinical Significance:

Variant	Therapy	
	In This Cancer	In Other Cancer
IIC TP53 p.(R175H)	▪ None*	▪ None*
APC c.1312+3A>G	▪ None*	▪ None*

Variant	Diagnostic Significance	Prognostic Significance
TP53 p.(R175H)	None*	None*
APC c.1312+3A>G	None*	None*

Biomarker Description:

Variant	Biomarker Description
APC NM_000038.6 chr5:112155044 c.1312+3A>G (Splice Site Mutation)	The APC gene encodes the adenomatous polyposis coli tumor suppressor protein that plays a crucial role in regulating the Î²-catenin/WNT signaling pathway which is involved in cell migration, adhesion, proliferation, and differentiation[30]. APC is an antagonist of WNT signaling as it targets Î²-catenin for proteasomal degradation[31,32]. Germline mutations in APC are predominantly inactivating and result in an autosomal dominant predisposition for familial adenomatous polyposis (FAP) which is characterized by numerous polyps in the intestine[30,33]. Acquiring a somatic mutation in APC is considered to be an early and possibly initiating event in colorectal cancer[34]. Alterations and prevalence Somatic mutations in APC are observed in up to 65% of colorectal cancer, and in up to 15% of stomach adenocarcinoma and uterine corpus endometrial carcinoma[6,7,35]. In colorectal cancer, ~60% of somatic APC mutations have been reported to

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	occur in a mutation cluster region (MCR) resulting in C-terminal protein truncation and APC inactivation[36,37]. Potential relevance Currently, no therapies are approved for APC aberrations.
TP53 NM_000546.6 chr17:7578406 c.524G>A p.(R175H)	The TP53 gene encodes the tumor suppressor protein p53, which binds to DNA and activates transcription in response to diverse cellular stresses to induce cell cycle arrest, apoptosis, or DNA repair[1]. In unstressed cells, TP53 is kept inactive by targeted degradation via MDM2, a substrate recognition factor for ubiquitin-dependent proteolysis[2]. Alterations in TP53 are required for oncogenesis as they result in loss of protein function and gain of transforming potential[3]. Germline mutations in TP53 are the underlying cause of Li-Fraumeni syndrome, a complex hereditary cancer predisposition disorder associated with early-onset cancers[4,5]. TP53 is the most frequently mutated gene in the cancer genome with approximately half of all cancers experiencing TP53 mutations. Ovarian, head and neck, esophageal, and lung squamous cancers have particularly high TP53 mutation rates (60-90%)[6,7,8,9,10,11]. Approximately two-thirds of TP53 mutations are missense mutations and several recurrent missense mutations are common, including substitutions at codons R158, R175, Y220, R248, R273, and R282[6,7]. Invariably, recurrent missense mutations in TP53 inactivate its ability to bind DNA and activate transcription of target genes[12,13,14,15]. Alterations in TP53 are also observed in pediatric cancers[6,7]. Somatic mutations are observed in 53% of non-Hodgkin lymphoma, 24% of soft tissue sarcoma, 19% of glioma, 13% of bone cancer, 9% of B-lymphoblastic leukemia/lymphoma, 4% of embryonal tumors, 3% of Wilms tumor and leukemia, 2% of T-lymphoblastic leukemia/lymphoma, and less than 1% of peripheral nervous system cancers (5 in 1158 cases)[6,7]. Biallelic loss of TP53 is observed in 10% of bone cancer, 2% of Wilms tumor, and less than 1% of B-lymphoblastic leukemia/lymphoma (2 in 731 cases) and leukemia (1 in 250 cases)[6,7]. The small molecule p53 reactivator, PC14586[16] (2020), received a fast track designation by the FDA for advanced tumors harboring a TP53 Y220C mutation. The FDA has granted fast track designation to the p53 reactivator, eprenetapopt[17], (2019) and breakthrough designation[18] (2020) in combination with azacitidine or azacitidine and venetoclax for acute myeloid leukemia patients (AML) and myelodysplastic syndrome (MDS) harboring a TP53 mutation, respectively. In addition to investigational therapies aimed at restoring wild-type TP53 activity, compounds that induce synthetic lethality are also under clinical evaluation[19,20]. TP53 mutation are a diagnostic marker of SHH-activated, TP53-mutant medulloblastoma[21]. TP53 mutations confer poor prognosis and poor risk in multiple blood cancers including AML, MDS, myeloproliferative neoplasms (MPN), and chronic lymphocytic leukemia (CLL), and acute lymphoblastic leukemia (ALL)[22,23,24,25,26,27]. In mantle cell lymphoma, TP53 mutations are associated with poor prognosis when treated with conventional therapy including hematopoietic cell transplant[28]. Mono- and bi-allelic mutations in TP53 confer unique characteristics in MDS, with multi-hit patients also experiencing associations with complex karyotype, few co-occurring mutations, and high-risk disease presentation as well as predicted death and leukemic transformation independent of the IPSS-R staging system[29].

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Therapy & Clinical Trials:

TP53 p.(R175H) c.524G>A

Therapy	FDA	NCCN	EMA	ESMO	Trials
atezolizumab	X	X	X	X	II
azenosertib	X	X	X	X	II
niraparib	X	X	X	X	II
ART-6043, olaparib, niraparib	X	X	X	X	I/II
ATRN-119	X	X	X	X	I/II
niraparib, GSK-101	X	X	X	X	I/II
VB-15010	X	X	X	X	I/II
CLSP-1025	X	X	X	X	I
GenSci-122	X	X	X	X	I
GH-2616	X	X	X	X	I
HRS-1167	X	X	X	X	I
HS-10502	X	X	X	X	I
NT-175, chemotherapy, aldesleukin	X	X	X	X	I
SC0245	X	X	X	X	I
SIM-0501	X	X	X	X	I
talazoparib, palbociclib, axitinib, crizotinib	X	X	X	X	I
TP53-EphA-2-CAR-DC, anti-PD-1	X	X	X	X	I
XL-309, olaparib	X	X	X	X	I

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APC c.1312+3A>G

Therapy	FDA	NCCN	EMA	ESMO	Trials
ST-316	X	X	X	X	I/II
tegatrabetan	X	X	X	X	I/II

? Note: ✓ : In this Cancer Type, ✓ : In Other Cancer Type, ✓✓ : both In this and Other Cancer Type, X : No Evidence.⚠?: Caution for Use

Test Description

Solidseq Comprehensive Panel is advancing precision oncology research through the analysis of multiple relevant biomarkers in a single next-generation sequencing (NGS) test.

The Solidseq Comprehensive Panel assay analyses **271** unique genes (**214** DNA + **57** RNA) for single gene and multiple gene biomarker insights, detects single gene biomarkers from all variant types including **SNVs, indels, CNVs, known and novel fusions, and splice variants**, analyze complex multi-gene biomarkers for mutational signatures, including microsatellite instability (**MSI**) for immunotherapy research.

The performance of the panel was verified using both commercially sourced reference controls and FFPE samples.

Test Methodology

Extraction & Library Preparation: DNA & RNA extracted from FFPE tissue block sample is used for targeted gene capture using a custom designed hybrid capture panel. In process quality controls were determined for prepared library.

Sequencing: The libraries were sequenced at mean depth: >1000x on Illumina Novaseq- next generation sequencing platform.

Data Analysis: Sequencing data are analyzed using a customized and validated bioinformatics pipeline capable of detecting various classes of genomic alterations, including single nucleotide variants (**SNVs**), **Indels**, copy number variations (**CNVs**) and fusions. As part of the streamlined bioinformatics workflow, Multi-gene Biomarkers such as Microsatellite Instability (**MSI**) Score is calculated. Reportable genomic alterations and fusions are prioritized, classified, and reported based on AMP-ASCO-CAP guidelines and NCCN guidelines.

Limit of Detection (LOD): The LOD for SNVs and InDels is 5% Variant allele Frequency (VAF) and for Fusions is ≥10 supporting reads.

Notes

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Treatment decisions should not be based solely on the results of this test. Findings must be interpreted in the context of the patient's complete clinical profile—including pathological, radiological, and family history—to guide diagnosis, prognosis, and therapeutic decisions.

Therapy-related information in this report is derived from multiple sources including FDA-approved drug data, NCCN guidelines, peer-reviewed literature, standard clinical databases, and biomarker evidence strength. However, therapy options listed may not be suitable or beneficial for all patients. This report aims to provide a summary of potentially effective treatments, therapies that may pose increased risks, and those that may have limited benefit, based on genomic alterations identified. Prognostic and diagnostic biomarkers relevant to the disease context may also be reported.

SN Genelab Pvt Ltd does not guarantee completeness, accuracy, or eligibility for trial enrollment. Physicians must verify eligibility criteria for specific trials independently.

Identification of a genomic alteration does not guarantee therapeutic efficacy or predict ineffectiveness. Final decisions on treatment should rest with the treating physician. The absence of reported therapeutic options does not imply ineffectiveness or lack of safety of any medication selected independently.

All variant classifications and related clinical information are based on the latest evidence from peer-reviewed publications, public clinical databases, and medical guidelines (e.g., NCCN, ASCO, AMP) available at the time of report generation. However, literature is continuously evolving, and classification may change with emerging evidence.

This test is performed on tumor tissue without a matched normal (e.g., blood) sample. Therefore, some variants may be of germline origin. While the assay is designed and validated for somatic variant detection, it does not differentiate between somatic and germline variants. If indicated, germline testing and genetic counselling are recommended.

FFPE-derived samples often pose challenges due to degradation. Poor-quality DNA or RNA may fail QC thresholds at various stages such as extraction, library preparation, or bioinformatics, and may result in assay failure or compromised results, such as low gene coverage or shallow variant depth.

Variants with allele frequencies between 3–5% may be reported if they are of strong or potential clinical relevance, or if previously detected in the same patient. False positives or negatives below assay sensitivity cannot be ruled out.

CNVs/gene amplifications have >80% detection sensitivity in this assay. Confirmation using orthogonal methods such as FISH/DDPCR is recommended when clinically indicated.

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Variants in non-coding regions or deep intronic sequences are not assessed in this assay.

This assay is under the scope of NABL as Solidseq Comprehensive panel.

DNA Gene list				
AKT1	AKT2	AKT3	ALK	APC
AR	ARAF	ARID1A	ARID5B	ATM
ATR	ATRX	AXL	BAP1	BARD1
BCOR	BCORL1	BLM	BRAF	BRCA1
BRCA2	BRIP1	BTK	CBL	CCND1
CCND2	CCND3	CCNE1	CDH1	CDK12
CDK2	CDK4	CDK6	CDKN1B	CDKN2A
CDKN2B	CHEK1	CHEK2	CIC	CREBBP
CSF1R	CSMD3	CTCF	CTNNB1	DDR2
DICER1	DPYD	EED	EGFR	EIF1AX
EP300	EPCAM	ERBB2	ERBB3	ERBB4
ERCC2	ESR1	EZH1	EZH2	FANCA
FANCC	FANCD2	FANCE	FANCF	FANCG
FANCI	FANCL	FARSB	FAT1	FAT3
FAT4	FBXW7	FGF19	FGF3	FGFR1
FGFR2	FGFR3	FGFR4	FLT3	FOXL2
FUBP1	GATA2	GATA3	GNA11	GNAQ
GNAS	H3F3A	HIST1H3B	HIST1H3C	HNF1A
HRAS	IDH1	IDH2	IGF1R	JAK1
JAK2	JAK3	KDM6A	KDR	KEAP1
KIT	KMT2C	KMT2D	KNSTRN	KRAS
LRP1B	MAGOH	MAP2K1	MAP2K2	MAP2K4
MAPK1	MAPK3	MAX	MDM2	MDM4
MED12	MEN1	MET	MLH1	MLH3
MPL	MRE11	MSH2	MSH3	MSH6
MTOR	MUC16	MUTYH	MYC	MYCL
MYCN	MYD88	NBN	NF1	NF2
NFE2L2	NOTCH1	NOTCH2	NOTCH3	NPM1
NRAS	NTRK1	NTRK2	NTRK3	PALB2
PBRM1	PDGFRA	PDGFRB	PICALM	PIK3CA
PIK3CB	PIK3CD	PIK3R1	PIK3R2	PMS1
PMS2	POLD1	POLE	PPARG	PPP2R1A
PPP2R2A	PTCH1	PTEN	PTH	PTPN11
RAC1	RAD50	RAD51	RAD51B	RAD51C
RAD51D	RAD52	RAD54L	RAF1	RB1

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RBM10	RET	RHEB	RHOA	RICTOR
RIT1	RNF213	RNF43	ROS1	SDHA
SDHB	SDHC	SDHD	SETD2	SF3B1
SLC5A5	SLX4	SMAD4	SMARCA4	SMARCB1
SMO	SPOP	SRC	STAT3	STK11
SUFU	SUZ12	SYN2	TERT	TG
TOP1	TP53	TSC1	TSC2	TSHR
U2AF1	VHL	XPO1	XRCC2	

RNA Gene list				
ABL1	AKT3	ALK	AXL	BRAF
CCND1	CIC	CLIP1	EGFR	EML4
ERBB2	ERBB4	ERG	ETV1	ETV4
ETV5	ETV6	EWSR1	FARSB	FGFR1
FGFR2	FGFR3	FGFR4	FUS	GLIS3
KIAA1549	KIF5B	MAML2	MET	MN1
MYB	MYBL1	NOTCH1	NOTCH2	NRG1
NTRK1	NTRK2	NTRK3	PAX8	PDGFB
PDGFRA	PDGFRB	PICALM	PPARG	PRKCA
RAF1	RET	RNF213	ROS1	SS18
STAT6	SYN2	TACC3	TFG	THADA
UACA	YAP1			

MSI	% of Unstable Loci
MSI - Stable	0-19%
MSI - Low / Intermediate	20-30%
MSI - High	>30%

Tier	Interpretation
Tier IA	Variants of Strong Clinical Significance: Biomarkers that predict response or resistance to US FDA-approved therapies for a specific type of tumor or have been included in professional guidelines as therapeutic, diagnostic, and/or prognostic biomarkers for specific types of tumors.
Tier IB	Variants of Strong Clinical Significance: Biomarkers that predict response or resistance to a therapy based on well-powered studies with consensus from experts in the field, or have diagnostic and/or prognostic significance of certain diseases based on well-powered studies with expert consensus.
Tier IIC	Variants of Potential Clinical Significance: biomarkers that predict response or resistance to therapies approved by FDA or professional societies for a different tumor type (i.e., off-label use of a drug), serve as inclusion criteria for clinical trials, or have diagnostic and/or prognostic significance

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	based on the results of multiple small studies.
Tier IID	Variants of Potential Clinical Significance: Biomarkers that show plausible therapeutic significance based on preclinical studies or may assist disease diagnosis and/or prognosis themselves or along with other biomarkers based on small studies or multiple case reports with no consensus.
Tier III	Variants of Unknown clinical significance (VUS): Not observed at a significant allele frequency in the general or specific subpopulation or pancancer or tumor specific variant databases. No convincing published evidence of cancer Association

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------------------ End Of Report ------------------

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Sample Cancer Type: Esophageal Cancer

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Relevant Biomarkers

Tier	Genomic Alteration	Relevant Therapies (In this cancer type)	Relevant Therapies (In other cancer type)	Clinical Trials
IIC	*MYC amplification*	None*	None*	5

*** Public data sources included in relevant therapies: FDA**1, NCCN, **EMA**2, ESMO
Tier Reference: Li et al. *Standards and Guidelines for the Interpretation and Reporting of Sequence Variants in Cancer: A Joint Consensus Recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists.* J Mol Diagn. 2017 Jan;19(1):4-23.

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Disclaimer: The data presented here are from a curated knowledge base of publicly available information, but may not be exhaustive. The data version is 2025.06(006). The content of this report has not been evaluated or approved by the FDA, EMA or other regulatory agencies.

Relevant Therapy Summary

● In this cancer type ○ In other cancer type ◐ In this cancer type and other cancer types ✖ No evidence

MYC amplification

Relevant Therapy	FDA	NCCN	EMA	ESMO	Clinical Trials*
entinostat, nivolumab	✖	✖	✖	✖	● (I/II)
ST-316	✖	✖	✖	✖	● (I/II)
tegatrabetan	✖	✖	✖	✖	● (I/II)
nedisertib, tuvusertib	✖	✖	✖	✖	● (I)
talazoparib, palbociclib	✖	✖	✖	✖	● (I)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Clinical Trials Summary

MYC amplification

NCT ID	Title	Phase
NCT03838042	INFORM2 Exploratory Multinational Phase I/II Combination Study of Nivolumab and Entinostat in Children and Adolescents With Refractory High-risk Malignancies (INFORM2-NivEnt)	I/II
NCT05687136	A Molecularly Driven Phase Ib Dose Escalation and Dose Expansion Study of the DNA-PK Inhibitor Peposertib (M3814) in Combination With the ATR Inhibitor M1774	I
NCT04693468	Modular Phase IB Hypothesis-Testing, Biomarker-Driven, Talazoparib Combination Trial (TalaCom)	I
NCT05848739	A Phase I-II Dose-escalation and Expansion Study of ST316 in Subjects With Selected Advanced Unresectable and Metastatic Solid Tumors	I/II
NCT04851119	A Phase I/II Study of Tegavivint (NSC#826393) in Children, Adolescents, and Young Adults With Recurrent or Refractory Solid Tumors, Including Lymphomas and Desmoid Tumors	I/II

Biomarker Descriptions

MYC amplification

MYC proto-oncogene, bHLH transcription factor

Background: The MYC gene encodes the MYC proto-oncogene, bHLH transcription factor (c-MYC), a basic helix-loop-helix transcription factor that regulates the expression of numerous genes that control cell cycle progression, apoptosis, metabolic pathways, and cellular transformation[1,2,3,4]. MYC is part of the MYC oncogene family, which includes the related transcription factors, MYCN and MYCL, and regulates transcription in 10-15% of promoter regions[5]. MYC functions as a heterodimer in complex with the transcription factor MAX[2,6].

Alterations and prevalence: Recurrent somatic alterations are observed in both solid and hematological cancers. Recurrent somatic mutations in MYC, including those at codon T58, are infrequent and hypothesized to increase the stability of the MYC protein[7,8]. Amplification of the MYC gene is observed in 15-30% of ovarian serous cystadenocarcinoma, esophageal adenocarcinoma, uterine carcinosarcoma, and breast invasive carcinoma, 10-15% of pancreatic adenocarcinoma, stomach adenocarcinoma, and liver hepatocellular carcinoma, 5-10% of head and neck squamous cell carcinoma, uterine corpus endometrial carcinoma, prostate adenocarcinoma, lung adenocarcinoma, lung squamous cell carcinoma, bladder urothelial carcinoma, and colorectal adenocarcinoma, and 2-5% of skin cutaneous melanoma, brain lower grade glioma, sarcoma, cervical squamous cell carcinoma, uveal melanoma, diffuse B-cell lymphoma, glioblastoma, and kidney chromophobe[9,10]. MYC is the target of the t(8;14)(q24;32) chromosomal translocation in Burkitt lymphoma that places MYC coding sequences adjacent to immunoglobulin region regulatory sequences, resulting in increased MYC expression[11,12]. Overall, MYC translocations are observed in 2% of diffuse large B-cell lymphoma[9,10]. Somatic mutations in MYC are observed in 7% of diffuse large B-cell lymphoma, 4% of uterine carcinosarcoma, 3% of uterine corpus endometrial carcinoma and skin cutaneous melanoma, and 2% of colorectal adenocarcinoma and stomach adenocarcinoma[9,10]. Alterations in MYC are also observed in pediatric cancers[10]. Somatic mutations in MYC have been observed in 59% of non-Hodgkin lymphoma, 2% of leukemia, and less than 1% of bone cancer (2 in 327 cases) and B-lymphoblastic leukemia/lymphoma (1 in 252 cases)[10]. Amplification of MYC is observed in 6% of embryonal tumor, 5% of bone cancer, and less than 1% of B-lymphoblastic leukemia/lymphoma (2 in 731 cases) and MYC translocations are observed in 5% of T-lymphoblastic leukemia/lymphoma[10].

Potential relevance: B-cell lymphoma with MYC translocations that co-occur with BCL2 or BCL6 are referred to as double hit lymphoma, while co-occurrence with BCL2 and BCL6 rearrangements is referred to as triple-hit lymphoma[13,14]. MYC translocations are a diagnostic marker of Burkitt Lymphoma[15,16]. MYC translocations are also indicative of high risk for multiple myeloma and are associated with poor risk in acute lymphoblastic leukemia[17,18]. Currently, no therapies are approved for MYC aberrations. Due to the high frequency of somatic MYC alterations in cancer, many approaches are being investigated in clinical trials including strategies to disrupt complex formation with MAX, including inhibition of MYC expression and synthetic lethality associated with MYC overexpression[1,19,20,21].

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